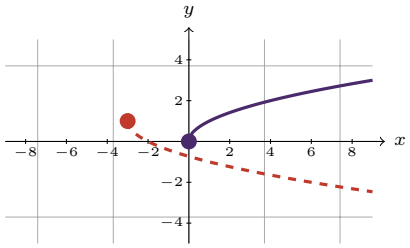
Instructions: Show all work in the space provided. Write each final answer in its box. Only what is inside a box is graded. Write the letter of the single best answer in its Letter box. Write intervals in interval notation. No notes. **70 points total.**

Part A — Transformations (12 pts)

1. The solid graph is $f(x) = \sqrt{x}$. The dashed graph is a transformation of f. Fill in the boxes so the equation matches the dashed graph.

$g(x) = \square\, f(\square\, x + \square) + \square$



2. The graph of $y = 3\sqrt{x-2}$ is the graph of the parent $y = \sqrt{x}$ after transformations. Complete the table: where does each landmark point of the parent land?

Parent point on $y = \sqrt{x}$	(0, 0)	(1, 1)	(4, 2)
Lands at on $y = 3\sqrt{x-2}$			

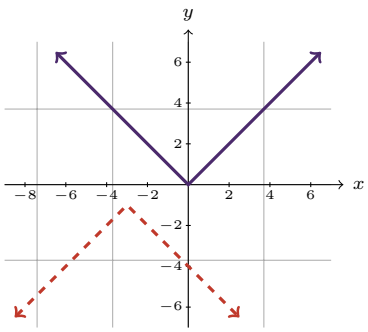
3. The graph of $y = \left|\frac{1}{4}x\right| - 3$ is the graph of the parent $y = |x|$ after transformations. Complete the table.

Parent point on $y = \lvert x \rvert$	$(-1, 1)$	(0, 0)	(1, 1)
Lands at on $y = \left\lvert\frac{1}{4}x\right\rvert - 3$			

4. The solid graph is $y = |x|$. Which function is the dashed graph?

(A) $y = -|x-3| - 1$
(B) $y = |x+3| - 1$
(C) $y = -|x+3| - 1$
(D) $y = -|x+1| - 3$

Letter: $\square$



Part B — Symmetry, One-to-One, and Inverses (14 pts)

5. Fill in each table of values. Then write whether the function is *even*, *odd*, or *neither*.

$f(x) = x^3 - 3x$

x	y
-2	
-1	
0	
1	
2	

$g(x) = 2|x| - 1$

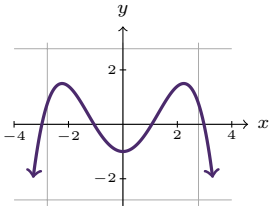
x	y
-2	
-1	
0	
1	
2	

$h(x) = x^2 + x$

x	y
-2	
-1	
0	
1	
2	

For 6–8, choose from the answers under each graph.

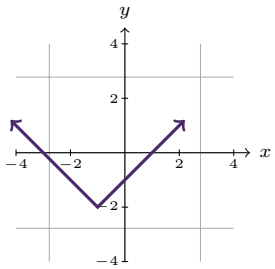
6. Is the function even, odd, or neither?



(A) even (B) odd (C) neither

Letter:

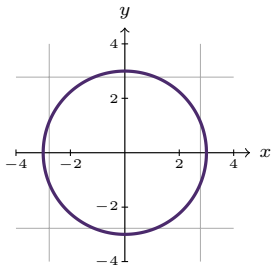
7. Use the vertical and horizontal line tests. The graph shows



(A) a one-to-one function
(B) a function, not one-to-one
(C) not a function

Letter:

8. Use the vertical and horizontal line tests. The graph shows



(A) a one-to-one function
(B) a function, not one-to-one
(C) not a function

Letter:

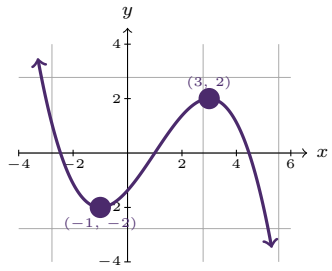
9. Let $f(x) = \sqrt[3]{3x+1}$. Which function is $f^{-1}(x)$?

(A) $\dfrac{x^3+1}{3}$ (B) $\dfrac{x^3}{3} - 1$ (C) $\dfrac{x^3-1}{3}$ (D) $\dfrac{1}{\sqrt[3]{3x+1}}$

Letter:

Part C — Reading Characteristics (21 pts)

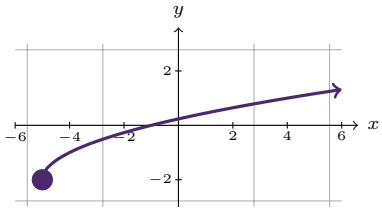
10. The graph of f continues in the direction of its arrows. Its turning points are marked.



Increasing on:

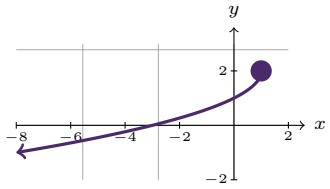
Decreasing on:

11. State the domain of the function graphed.



Domain:

12. State the range of the function graphed.

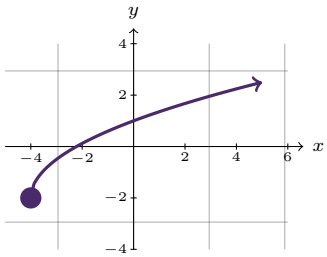


Range:

13. Find the domain of $f(x) = \sqrt{3x + 12} - 4$ algebraically.

Domain:

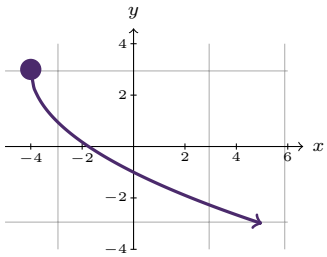
14. Is the graph concave up or concave down?



(A) concave up (B) concave down

Letter:

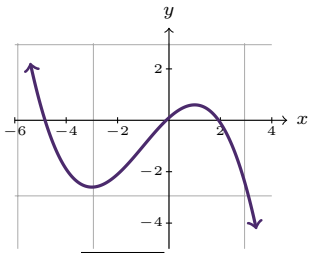
15. Is the graph concave up or concave down?



(A) concave up (B) concave down

Letter:

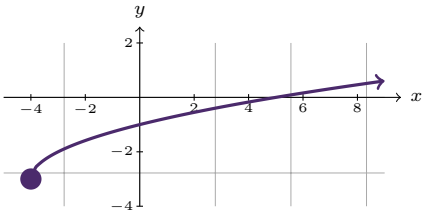
16. Find the point(s) of inflection.



How many?

At:

17. The function f is graphed at the right.



(a) Domain of f: ☐

(b) Range of f: ☐

(c) Which sentence is true?

(A) f is not invertible because it is not one-to-one.

(B) f is invertible because it passes the vertical line test.

(C) f is invertible because it passes the horizontal line test.

Letter: ☐

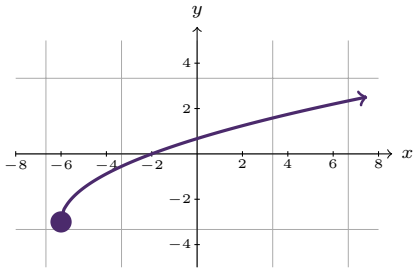
(d) Domain of f^{-1}: ☐

(e) Range of f^{-1}: ☐

18. Which type of function is graphed?

(A) exponential
(B) absolute value
(C) cube root
(D) square root
(E) linear

Letter: ☐



Part D — Average Rate of Change and Secant Lines (6 pts)

Let $h(x) = \sqrt[3]{x+1} - 2$.

19. Find the average rate of change of h from $x = -9$ to $x = 7$. Write your answer as a simplified fraction.

☐

20. Find the secant line of h from $x = -9$ to $x = 7$, in slope-intercept form. $y =$ [ ]

Part E — One Function, Every Characteristic (17 pts)

Problems 21–27 are all about the function

$$g(x) = 2\,|x + 1| - 6.$$

21. Describe the transformations that take $y = |x|$ to g.

[ ]

22. The domain of g is [ ], and the range is [ ].

23. g is increasing on [ ] and decreasing on [ ].

24. Find every intercept of g.

x-intercept(s): $x =$ [ ] y-intercept: [ ]

For 25 and 26, choose from these four statements.

(A) $\lim_{x\to\infty} g(x) = \infty$ (B) $\lim_{x\to-\infty} g(x) = \infty$ (C) $\lim_{x\to\infty} g(x) = -\infty$ (D) $\lim_{x\to-\infty} g(x) = -\infty$

25. The left end behavior of g: *Letter:* [ ] **26.** The right end behavior of g: *Letter:* [ ]

27. Complete the sentence: the [ ] (*maximum* or *minimum*) value of g is [ ], and it occurs at $x =$ [ ].